BIM-Based Construction Progress Monitoring Using Drone Simulation

Project Proposal

Problem

BIM represents intended construction state, not actual site progress. Manual inspection does not scale. **Question:** can drone-collected imagery automatically determine BIM component completion and overall progress?

Objectives

1. Use an IFC BIM model (walls, columns, beams, slabs, doors/windows) with unique component IDs as ground truth.
2. Generate a simulated incomplete state $BIM_{current} \subset BIM_{complete} = \{E_1, \ldots, E_n\}$ at controlled levels (25/50/75/90%).
3. Simulate a drone/camera $P = (x, y, z, \theta, \phi, \psi)$ under grid, perimeter, waypoint, or adaptive inspection.
4. Detect construction elements in captured images; localize in 3D.
5. Match detections to BIM component IDs by position, dimensions, spatial relationships.
6. Assign per-component status (not started / not detected / partial / completed / uncertain); aggregate to overall progress.

Methodology

1. **BIM Preparation** – parse IFC; extract IDs, categories, geometry, dimensions, relationships as ground truth.
2. **Incomplete Site Generation** – programmatically remove selected elements to produce controlled partial-completion states.
3. **Drone Simulation & Capture** – virtual drone captures RGB plus, in simulation, ground-truth depth, pose, and segmentation from multiple viewpoints.
4. **Element Detection** – Images → Detection/Segmentation → 3D Localization → BIM Matching. Initial scope: columns, beams, walls, slabs.
5. **Comparison & Scoring** – per-component completion $C_i \in [0, 1]$; overall progress

$$Progress = \frac{\sum_i w_i C_i}{\sum_i w_i} \times 100, \quad w_i = 1 \text{ initially, refined later}$$

Simulation Environment

Developed in simulation before real-site access. Stack: IfcOpenShell/Bonsai for IFC parsing and element removal; NVIDIA Isaac Sim + PegasusSimulator + Replicator, or Blender with scripted camera paths, for rendering, pose control, and ground-truth depth/segmentation. Known pose and depth in simulation allow direct 3D back-projection of detections – no structure-from-motion required at this stage.

Evaluation

Known ground truth (e.g. 70/100 components present) compared against system output.

- Accuracy = correctly identified components / total components
- Precision and recall of completed-component classification
- Progress error: $|Progress_{est} - Progress_{GT}|$
- Localization error of matched components; coverage: % of components visible from captured viewpoints

Scope of Initial Prototype

Major structural components only (columns, beams, walls, slabs); camera-pose simulation rather than full drone flight dynamics – concentrates effort on the core estimation problem before extending scope.

Extensions

Real UAV deployment with RGB/depth/LiDAR; automated BIM-derived flight planning; multi-view 3D reconstruction (SfM/MVS) for real-world capture without known pose; partial-completion detection; change detection vs. plan; time-series monitoring $t_1 \rightarrow t_2 \rightarrow t_3$; validation on a real construction site.

Significance

Bridges the static BIM model and the dynamic physical state of a site. Simulation-first evaluation provides controlled, known-ground-truth conditions to test trajectories, detection methods, and matching techniques before real-world deployment:

BIM → Drone Inspection → Visual Understanding → BIM Matching → Progress Estimation